

Similarity and normalisation

Project: Image-pattern Network Investigation

1. Prepare the purpose, prerequisites and materials

Teach the learning cycle

Primary target: Calculate vector length and normalise a nonzero vector

Session · Sequence and minutes · Resources

Session: Session 1 · Understand · 45 min

Sequence and minutes: 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.

Resources: Sections 1–6; record the brief recall in the workbook.

Session: Session 2 · Practise · 45 min

Sequence and minutes: 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.

Resources: Sections 7–10; use the matching workbook W2 and misconception check.

Session: Session 3 · Investigate and transfer · 45 min

Sequence and minutes: 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.

Resources: Sections 9 and 11–14. Give extra time to finish the full task set; use Section 15 when the week includes a coding lab.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building. Repeated copies of a case share one recorded attempt; they are not new evidence.

Retrieval answer guidance

Embeddings and neighbourhoods

Without opening the old lesson, explain one key idea from Embeddings and neighbourhoods. Give an example and a mistake that the idea helps you avoid.

Look for: Look up a vector using a token ID and preserve sequence shape

Reference example, not the only acceptable wording: Teacher explanation: Vectors are $\begin{bmatrix} 1 & 1 \\ 0 & 1 \\ 0 & 2 \\ 0 & 0 \end{bmatrix}$, mask $[1, 1, 1, 0]$ and shape 4 by 2.

Augmentation and leakage

Without opening the old lesson, explain one key idea from Augmentation and leakage. Give an example and a mistake that the idea helps you avoid.

Look for: Check whether a transformation preserves the task label

Reference example, not the only acceptable wording: Teacher explanation: Original $d = -2$ and class 0. Row swapping leaves $d = -2$ and class 0. Column swapping gives rows $[1, 0]$, $[1, 0]$, $d = 2$ and class 1.

Convolution as a local operation

Without opening the old lesson, explain one key idea from Convolution as a local operation. Give an example and a mistake that the idea helps you avoid.

Look for: Compute a local weighted sum at each valid image position

Reference example, not the only acceptable wording: Teacher explanation: Window rows are $[2, 0]$ and $[1, 3]$. Products are 2, 0, 0 and 3, giving output 5.

Use two 45-minute sessions. Supply calculators and connect vector length to the right-triangle calculation 3 squared plus 4 squared equals 5 squared. Students should justify a rule by the information it retains. The supplied vectors are constructed examples; no semantic or accuracy claim is established by their geometry.

Outcomes

Calculate vector length and normalise a nonzero vector

Compare squared distance with cosine similarity and explain a changed ranking

Handle zero vectors explicitly and justify what information a comparison keeps

Preparation

Printed lesson and workbook; paper and pencil

Calculator and optional local Python 3 editor; no packages or AI accounts needed

Use the fictional examples supplied; prepare the fresh retry before class

Prerequisite answer

The squared distance is $(3 - 3)^2 + (4 - 0)^2 = 16$. It describes closeness under this representation and rule; it does not prove equivalent meaning.

End goal for the pictorial case

By the end, I can explain similarity and normalisation using a traced example and a stated limitation.

This added worked case illustrates the week's main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

2. Explain and model the mechanism

Last week, a neighbourhood depended on a chosen distance rule. This week, compare the same fictional vectors in two ways. Squared distance rewards nearby coordinates. Cosine similarity compares direction and ignores positive changes in length. Neither rule is automatically right for every task.

For Q, length = square root of $(3^2 + 4^2) = \text{square root of } 25 = 5$. Divide both components by 5 to get its unit vector $[0.6, 0.8]$. Its length is one because $0.6^2 + 0.8^2 = 1$. A becomes the same unit vector. C becomes $[-0.6, -0.8]$, pointing in the opposite direction.

These supplied coordinates are for arithmetic practice. They are not learned language embeddings. Normalising each row keeps its direction but removes its original length unless that length is saved separately.

Multiply matching components and add to obtain the dot product. Cosine similarity equals $\text{dot}(Q, \text{candidate}) / (\text{length}(Q) \times \text{length}(\text{candidate}))$. Higher cosine means a closer direction. For A, the dot product is $3 \times 6 + 4 \times 8 = 50$, so cosine = $50 / (5 \times 10) = 1$. For B it is $9 / 15 = 0.6$; for C it is $-25 / 25 = -1$.

Trace before running: A gives 1, B gives 0.6 and C gives -1. This classroom code assumes finite, moderate numeric inputs. It rejects empty, unequal-length and zero vectors. Keep full precision for decisions; rounding here is only for display. Cosine lies between -1 and 1 in exact arithmetic and is not a probability.

Term · Meaning

Term: Vector length

Meaning: The square root of the sum of squared components, also called the L2 norm

Term: L2 normalisation

Meaning: Dividing every component of a nonzero vector by its length

Term: Cosine similarity

Meaning: The dot product of two nonzero vectors divided by the product of their lengths

Term: Unit vector

Meaning: A vector with length one

Compare the same candidates using two rules

Candidate · Raw squared distance · Cosine similarity

Candidate: A [6, 8]

Raw squared distance: 25

Cosine similarity: 1

Candidate: B [3, 0]

Raw squared distance: 16

Cosine similarity: 0.6

Candidate: C [-3, -4]

Raw squared distance: 100

Cosine similarity: -1

For Q [3, 4], raw squared distance selects B by minimising the score. Cosine selects A by maximising the score. The candidates stayed the same; the comparison rule changed.

Guided: D has squared distance $9 + 1 = 10$ and cosine $20 / 25 = 0.8$. Raw distance prefers D over A (10 versus 25); cosine prefers A (1 versus 0.8). Exit: Q and A share direction but have different lengths and may represent different totals. Cosine 1 proves aligned nonzero vectors, not equivalent meaning. Check the task, retained information, zero and tie policies, and validation performance against a baseline; reserve final evaluation.

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a prediction, then explain the deciding step. These teacher notes contain solutions; do not reveal the independent case answer during modelling.

Compare the same candidates using two rules

Candidate · Raw squared distance · Cosine similarity

Candidate: A [6, 8]

Raw squared distance: 25

Cosine similarity: 1

Candidate: B [3, 0]

Raw squared distance: 16

Cosine similarity: 0.6

Candidate: C [-3, -4]

Raw squared distance: 100

Cosine similarity: -1

For Q [3, 4], raw squared distance selects B by minimising the score. Cosine selects A by maximising the score. The candidates stayed the same; the comparison rule changed.

After normalising both Q and every candidate, squared distances to A, B and C are 0, 0.8 and 4. For unit vectors, squared distance = $2 - 2 \times \text{cosine}$. Thus the smallest normalised distance and largest cosine produce the same ranking, including ties, in exact arithmetic.

Add candidate D [0, 5]. Calculate its raw squared distance and cosine from Q [3, 4]. Compare D with A under each rule.

Thinking move · Teacher prompt

Thinking move: Plan

Teacher prompt: What is given? What is the question? Which procedure fits these facts?

Thinking move: Do

Teacher prompt: Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.

Thinking move: Check

Teacher prompt: Does the output answer the question? Which assumption or limitation prevents a larger claim?

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: Similarity and normalisation

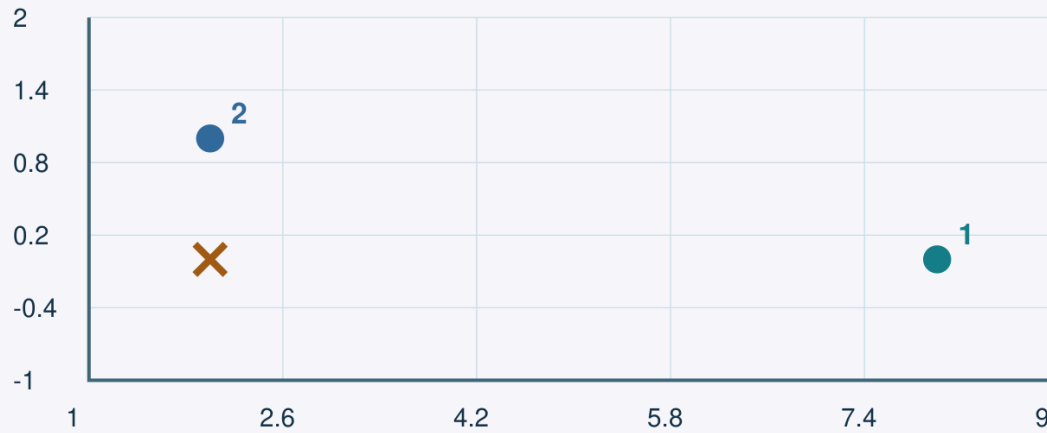
Compare angular alignment with raw distance on the same vectors.

The facts we will use

Query $[2, 0]$; $P=[8, 0]$; $Q=[2, 1]$. Nonzero vectors only; compare squared Euclidean distance and cosine.

All inputs and outcomes in this worked example are invented classroom data; no real model run is claimed.

Raw proximity differs from angular alignment



Horizontal: first vector coordinate; vertical: second vector coordinate

1: P at (8, 0)

2: Q at (2, 1)

Cross: query at (2, 0)

Distance rule: Compare squared Euclidean distance and cosine

Similarity and normalisation

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Raw distances

P distance 36; Q distance 1, so Q is closer in raw coordinates.



2. Angular similarity

$\text{Cosine}P=1$; $\text{cosine}Q=2/\sqrt{5}\approx 0.8944$, so P aligns more closely.



3. Normalise

Unit query $[1, 0]$, P $[1, 0]$, Q $[2/\sqrt{5}, 1/\sqrt{5}]$; normalised distance P0.

CHANGE ONE THING

Replace P with $[0, 0]$.

Explain the changed result using the matching lesson.

Read the picture one step at a time

Step · What happens and why

Step: 1. Raw distances

What happens and why: P distance 36; Q distance 1, so Q is closer in raw coordinates.

Step: 2. Angular similarity

What happens and why: $\text{Cosine}P=1$; $\text{cosine}Q=2/\sqrt{5}\approx 0.8944$, so P aligns more closely.

Step: 3. Normalise

What happens and why: Unit query $[1, 0]$, $P [1, 0]$, $Q [2/\sqrt{5}, 1/\sqrt{5}]$; normalised distance PQ .

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Raw Euclidean distance includes differences in both direction and magnitude. Cosine compares direction using the vector lengths, so a long vector can align perfectly while being far away in raw coordinates. Declare which question your similarity measure answers. A zero vector has no direction and cannot be divided by its norm for cosine or unit normalisation.

What this example does not establish

Neither metric is universally right; choose it from the task and handle zero vectors explicitly.

Change one thing in the pictured case

Change: Replace P with $[0, 0]$.

Prediction: Cosine and unit normalisation for P are undefined.

Reason: The zero vector has norm 0, so dividing by its norm is invalid.

Ask while showing the picture

Cover the result. Ask learners to name the inputs and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its evidence. Then change the supplied condition and compare. The picture is a representation; it does not document execution of a real AI model.

3. Demonstrate and guide practice

Time or stage · Teacher action

Time or stage: Session 1 start to 7 minutes

Teacher action: Retrieve squared distance and explain why a geometric score alone does not establish meaning.

Time or stage: 7 to 18 minutes

Teacher action: Compute lengths and unit vectors. Complete W1 with both positive and opposite directions.

Time or stage: 18 to 32 minutes

Teacher action: Trace the cosine formula and optional code, checking which score is maximised.

Time or stage: 32 to 45 minutes

Teacher action: Use the diagram and W2 to contrast raw and normalised distance rankings.

Time or stage: Session 2 start to 8 minutes

Teacher action: Complete the guided D comparison, retaining the same query and candidate A.

Time or stage: 8 to 20 minutes

Teacher action: Discuss event counts and zero vectors. Complete W3 and require an explicit no-comparison outcome.

Time or stage: 20 to 35 minutes

Teacher action: Collect W4 independently, including raw scores, cosines, normalised distances and an explanation.

Time or stage: 35 to 45 minutes

Teacher action: Complete W5 and the exit response. Give feedback before administering the fresh retry.

Use the worked example in Sections 3 and 6. Collect a prediction before revealing its result; ask the learner to point to each intermediate fact.

Provide a ledger for squared components, length, unit components and score. Allow a calculator and separate lower-is-better distance from higher-is-better similarity. Extension: expand the squared distance between two unit vectors to derive $2 - 2 \times \text{their dot product}$. For floating-point retrieval, declare a tolerance and retain all scores within it of the best; show that rounding first can create false ties.

Reduce help deliberately

Completion task: For query Q, write the raw squared distances and cosines to A, B and C from the lesson. Name the winner under each rule. Calculate squared distances after normalising both sides and check the relationship to cosine.

Teacher check: Teacher explanation: Raw squared distances are 25, 16, 100, so B wins. Cosines are 1, 0.6, -1, so A wins. Unit-vector squared distances are 0, 0.8, 4, matching $2 - 2 \times \text{each cosine}$ and selecting A.

Start by identifying the given inputs together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood. For partners, alternate explainer and checker; collect a response from every learner before discussion.

Individual reasoning check

Use the worked case for Similarity and normalisation. Proposed explanation: “Cosine is a probability”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

Expected correction: Its possible negative values and geometric definition show that it is not a calibrated probability.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class. If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Pictorial practice question

Why do the original two metrics select different winners? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.

Reasoned teacher answer

Raw distance includes length differences; cosine compares direction after accounting for norms. P shares direction but has a much larger magnitude. Changed-case check: Cosine and unit normalisation for P are undefined. The zero vector has norm 0, so dividing by its norm is invalid.

Changed-case reasoning: Cosine and unit normalisation for P are undefined. The zero vector has norm 0, so dividing by its norm is invalid.

Collect each learner’s explanation before discussion. Use the first missing connection between input, deciding step and result to choose support. This question is worked-case practice, not fresh mastery evidence.

4. Run the investigation and diagnose misconceptions

Cosine is a probability: Its possible negative values and geometric definition show that it is not a calibrated probability.

Normalisation keeps all information: Positive multiples collapse to the same unit vector; preserve length separately if the task needs it.

Any scaling leaves cosine unchanged: Positive scaling preserves direction. Multiplying only one vector by a negative number reverses the cosine sign; zero scaling makes it undefined.

Use Section 9 for the evidence ledger. Add a boundary or missing-input case and record the expected response before checking. More successful repeats of one case do not establish wider coverage.

Match the support to the first error

What the response shows · Next teaching action

What the response shows: Missing or misread facts

Next teaching action: Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.

What the response shows: Wrong procedure

Next teaching action: Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.

What the response shows: Arithmetic or implementation error

Next teaching action: Trace intermediate values or lines; preserve the first mismatch and retry that step.

What the response shows: Correct result without a reason

Next teaching action: Ask for a trace and a changed case before marking independent understanding.

What the response shows: An unsupported wider claim

Next teaching action: Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal. Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

5. Assess and explain the answers

Guided and exit questions

Guided: D has squared distance $9 + 1 = 10$ and cosine $20 / 25 = 0.8$. Raw distance prefers D over A (10 versus 25); cosine prefers A (1 versus 0.8). Exit: Q and A share direction but have different lengths and may represent different totals. Cosine 1 proves aligned nonzero vectors, not equivalent meaning. Check the task, retained information, zero and tie policies, and validation performance against a baseline; reserve final evaluation.

W1 Trace the unit vectors

For Q [3, 4], A [6, 8] and C [-3, -4], give each length and unit vector. Explain one distinction that normalisation removes and one that it preserves.

Teacher explanation: Lengths are 5, 10 and 5. Unit vectors are [0.6, 0.8], [0.6, 0.8] and [-0.6, -0.8]. The length difference between Q and A disappears; the opposite direction of C is preserved.

W2 Compare the rankings

For query Q, write the raw squared distances and cosines to A, B and C from the lesson. Name the winner under each rule. Calculate squared distances after normalising both sides and check the relationship to cosine.

Teacher explanation: Raw squared distances are 25, 16, 100, so B wins. Cosines are 1, 0.6, -1, so A wins. Unit-vector squared distances are 0, 0.8, 4, matching 2 - 2 times each cosine and selecting A.

W3 Handle a vector without direction

A candidate set contains Z [0, 0] and B [3, 0]. With nonzero query Q, what happens under this lesson's zero-vector policy? What if Q is zero, or Z is the only candidate? Explain why replacing an undefined cosine with zero is misleading.

Teacher explanation: Exclude Z and report it; compare B normally. Reject a zero query. If only Z exists, report no comparison. Zero cosine represents perpendicular nonzero directions, whereas the zero vector has no direction.

W4 Compare a new case independently

Use query [5, 0] and candidates P [10, 0], R [4, 3] and S [0, 5]. Calculate all raw squared distances and cosines, name each winner, and give squared distances after normalising both query and candidates. Explain the changed ranking.

Teacher explanation: Raw squared distances to P, R and S are 25, 10, 50, selecting R. Cosines are 1, 0.8, 0, selecting P. Normalised squared distances are 0, 0.4, 2. P shares the query direction despite its greater length; raw distance also reflects that length difference.

W5 Audit a proposed improvement

For event-count vectors [3, 4] and [6, 8], what useful information can normalisation remove? A team chooses cosine because it improves results on the final evaluation set, then calls that same set independent evidence. Explain how to repair the evaluation plan.

Teacher explanation: Normalisation removes magnitude: totals 7 and 14 become indistinguishable in the unit vectors. Choose the rule using task needs and training/validation evidence, preserving a baseline. A final set used to choose the rule is no longer independent final evidence; reserve new untouched evaluation data or explicitly report the limitation.

Independent assessment and fresh retry

W4 requires raw scores 25, 10, 50; cosines 1, 0.8, 0; normalised distances 0, 0.4, 2; and winners R then P. Fresh retry: query [0, 5], candidates D [0, 10], E [3, 4], F [5, 0]. Raw squared distances are 25, 10, 50; cosines 1, 0.8, 0; unit distances 0, 0.4, 2. Raw distance selects E; cosine selects D. Ask students to explain why a zero query is rejected.

Assess independent transfer

Selected case: Use query [5, 0] and candidates P [10, 0], R [4, 3] and S [0, 5]. Calculate all raw squared distances and cosines, name each winner, and give squared distances after normalising both query and candidates. Explain the changed ranking.

Selected case answer

Teacher explanation: Raw squared distances to P, R and S are 25, 10, 50, selecting R. Cosines are 1, 0.8, 0, selecting P. Normalised squared distances are 0, 0.4, 2. P shares the query direction despite its greater length; raw distance also reflects that length difference.

Dimension · Evidence required

Dimension: Explanation

Evidence required: Names the applicable idea or procedure and ties it to the supplied facts.

Dimension: Trace and result

Evidence required: Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.

Dimension: Limits

Evidence required: States the assumptions, missing evidence or conditions under which the result would change.

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace. Do not penalise an honestly recorded model failure if the learner correctly explains its evidence and limits.

Feedback that the learner uses

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case. Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

6. Connect the project, support and next lesson

Add a labelled similarity and normalisation evidence sheet to Image-pattern Network Investigation. Show how you can calculate vector length and normalise a nonzero vector. Use the supplied fictional case and keep its inputs, intermediate steps, calculation or prediction, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

Provide a ledger for squared components, length, unit components and score. Allow a calculator and separate lower-is-better distance from higher-is-better similarity. Extension: expand the squared distance between two unit vectors to derive $2 - 2 \text{ times their dot product}$. For floating-point retrieval, declare a tolerance and retain all scores within it of the best; show that rounding first can create false ties.

Week 28 examines transfer learning, comparing a reused representation with a baseline and testing what can transfer to a new task.

Retain the project ledger and assess a changed case after support. Check the relevant prerequisite before carrying this component forward.

Plan the delayed return

Suggested return points in this level: ai-5/28, ai-5/30, ai-5/33. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used. For a new level, revisit the previous capstone before checking the new prerequisites.

Record · Teacher evidence

Record: Learner code and date

Teacher evidence: _____

Record: Concept and case used; prior exposure

Teacher evidence: _____

Record: Explanation / trace / limits

Teacher evidence: _____

Record: Support and first repair

Teacher evidence: _____

Record: Changed case and delayed outcome

Teacher evidence: _____

Record: Next teaching action

Teacher evidence: _____

Use the evidence to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

Project evidence from this case

For your Image-pattern Network Investigation, save the input, procedure, observed result and limitation of this check. Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.